ACADEMIC CITY UNIVERSITY

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TOPIC :

DEVELOPMENT OF AN AUDITABLE DUAL-EXPERT MLOPS FRAMEWORK FOR DIABETIC RETINOPATHY AND GLAUCOMA SCREENING: A DISTRIBUTED SYSTEM APPROACH FOR HEALTHCARE

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LIST OF ACRONYMS

Clinical & Diagnostic Terms

DR: Diabetic Retinopathy

CDR: Cup-to-Disc Ratio (Key metric for Glaucoma assessment)

OD / OS: Oculus Dexter (Right Eye) / Oculus Sinister (Left Eye)

Grad-CAM: Gradient-weighted Class Activation Mapping (Explainability technique)

HL7 / FHIR: Health Level Seven / Fast Healthcare Interoperability Resources (Medical data standards)

Artificial Intelligence & Frameworks

CNN: Convolutional Neural Network

EfficientNet: Efficient Network (High-performance backbone architecture)

PyTorch: Tensors and Dynamic neural networks in Python (Used for the Glaucoma Expert Model)

TensorFlow / Keras: Open-source library for Machine Learning (Used for the DR Expert Model)

TL: Transfer Learning

Focal Loss: Loss function designed to handle extreme foreground-background class imbalance

MLOps & Data Engineering

MLOps: Machine Learning Operations

Kafka: Apache Kafka (Distributed event store and stream-processing platform)

PostgreSQL: Open-source Relational Database Management System (Metadata Store)

S3 / MinIO: Object Storage Service (Used for clinical image persistence)

MLflow: Platform for the Machine Learning lifecycle (Tracking and Registry)

API: Application Programming Interface

CI/CD: Continuous Integration / Continuous Deployment

ABSTRACT

This study proposes OculoGuard (v2.0), an advanced MLOps framework designed for the robust screening of Diabetic Retinopathy (DR) and Glaucoma in resource-constrained environments. Unlike standard monolithic architectures, OculoGuard implements a Dual-Expert Model strategy: two specialized neural networks trained independently to achieve clinical-grade precision.

The first expert utilizes a fine-tuned EfficientNet-B3 backbone for high-sensitivity DR Classification (Stages 0-4), while the second expert focuses on CDR Regression (Cup-to-Disc Ratio) for glaucoma assessment. To ensure clinical trust, both models integrate Grad-CAM explainability maps, allowing clinicians to visualize the pathological features driving the AI's decision.

Beyond diagnostic accuracy, the framework addresses system resilience and data integrity through a production-grade MLOps loop. We implement an asynchronous metadata pipeline powered by Apache Kafka, which decouples the inference engine from the storage layer. This ensures that even under high load, every scan, prediction, and confidence score is reliably streamed and recorded incrementally into a PostgreSQL Database. Clinical images and model versions are managed via S3/MinIO and MLflow Registry, providing full lineage and rollback capabilities.

By bridging the gap between deep learning and distributed systems engineering, OculoGuard provides a scalable, auditable, and transparent AI solution ready for integration into Ghana's healthcare infrastructure.

I. INTRODUCTION

Since the beginning of the 21st century, the digital transformation of healthcare systems has radically redefined the global clinical landscape, placing high-resolution medical imaging and rapid diagnostic flows at the heart of public health strategies. While this transition has promoted universal health coverage and early disease detection, it has also highlighted a critical shortage of specialized medical expertise, particularly in developing economies. In 2026, the automated screening of ophthalmic pathologies is no longer just a technical challenge for research laboratories, but a matter of national health security and social stability in Ghana. According to World Health Organization (WHO) projections, the prevalence of vision loss due to unmanaged chronic conditions is expected to double by 2030, highlighting the urgent need for robust, scalable, and auditable diagnostic systems.

The dynamics of securing clinical diagnostic flows play a crucial role in epidemiological stability. From a theoretical perspective, this issue falls within the scope of computer vision (Marr, 1982) and medical decision theory (Hunink et al., 2014). The asymmetry of information between the limited number of ophthalmologists and the growing volume of diabetic patients creates a diagnostic bottleneck, where the manual identification of subtle microaneurysms or Cup-to-Disc Ratio (CDR) variations becomes prone to human fatigue. As observed in the foundational work of **Gulshan et al. (2016)**, the efficiency of mass screening programs depends on reducing false negative rates and optimizing referral transaction costs. An infrastructure unable to provide explainable feedback in real-time suffers an erosion of clinical trust, thereby hindering the adoption of digital health solutions.

Historically, ophthalmic screening relied on manual grading by experts based on rigid clinical scales (e.g., the ETDRS scale). However, these methods have shown their limitations in the face of the massive volume of data generated by modern fundus cameras. While classical image processing and statistical pattern recognition provided important foundations, they struggle to capture the complex non-linear features and high-dimensional spatial hierarchies inherent in retinal pathologies. It is in this context that deep learning methods offer new perspectives. According to **LeCun et al. (2015)**, these architectures make it possible to process raw pixel data and capture

intricate biological patterns, surpassing traditional feature engineering in terms of diagnostic sensitivity (**Ting et al., 2017**).

Nevertheless, the application of Artificial Intelligence (AI) to this field faces a major operational obstacle: the "Black-Box" nature of deep neural networks and the volatility of clinical data streams. In real-world hospital environments, AI models often operate in isolation, lacking the transparency required for regulatory compliance. In order to address this structural constraint, this work employs a **Dual-Expert approach** combining a **TensorFlow-based** model for Diabetic Retinopathy (DR) classification with a **PyTorch-based** expert for Glaucoma assessment (CDR). Performance evaluation is primarily based on sensitivity and the F1-score, metrics particularly suited to clinical environments where missing a positive case (false negative) carries a high human cost.

The major innovation of this project, called 'OculoGuard v2.0', lies in moving beyond static inference models by designing a dynamic, distributed framework. This architecture integrates an Apache Kafka-based asynchronous streaming layer and a PostgreSQL-based metadata store. This infrastructure enables the continuous recording of every diagnostic transaction, feeding an incremental Audit Trail designed to facilitate clinical monitoring and system transparency. This approach aims to ensure continuous adaptation to diverse clinical datasets over the 2024-2026 period, while maintaining the infrastructural resilience of the system.

Finally, in order to ensure the transparency essential for medical ethics and algorithmic governance, we are integrating Explainable Artificial Intelligence (XAI) mechanisms via Grad-CAM visualizations. This approach transforms deep learning models into interpretable decision-making tools, enabling clinicians to identify the specific pathological regions—such as exudates or optic disc excavations—that determine the classification of a scan. By reconciling algorithmic power, dual-expert specialization, infrastructural resilience through asynchronous streaming, and clinical interpretability, this project aims to provide an integrated framework for retinal screening that is adapted to the contemporary requirements of the digital health economy .

II. PROBLEM STATEMENT AND RESEARCH GAP

The screening of ocular diseases in Ghana faces a triple challenge that current academic models fail to address simultaneously:

Specialist Scarcity and Diagnostic Delay: With a critically low ophthalmologist-to-patient ratio, manual screening is unsustainable. Existing AI research focuses on accuracy but ignores the high-throughput requirement of national screening programs.

Architectural Fragility (The MLOps Gap): Most contemporary studies (2020–2025) produce "static models." These models lack an asynchronous layer (like Apache Kafka) to handle concurrent data streams from multiple regional clinics, leading to system latency or failure under load.

The "Black-Box" Accountability Crisis: There is a significant lack of Auditable Metadata Stores. Without a persistent, incremental record (like a PostgreSQL-based Audit Trail), AI-driven diagnoses remain legally and clinically difficult to validate in a formal medical environment.

III. RESEARCH OBJECTIVES

3.1 General Objective

The primary objective of this study is to develop and deploy OculoGuard v2.0, an integrated MLOps framework featuring a Dual-Expert System for the automated, resilient, and auditable screening of Diabetic Retinopathy and Glaucoma.

3.2 Specific Objectives

- Design a Dual-Expert Architecture: Implementing a TensorFlow-based model for DR classification and a PyTorch-based model for CDR regression to maximize diagnostic sensitivity.

- Build an Asynchronous Data Pipeline: Leveraging Apache Kafka to ensure high-throughput image processing and system resilience across distributed clinical nodes.
- Implement a Metadata Audit Trail: Developing a PostgreSQL storage layer to record every inference transaction, ensuring clinical traceability and regulatory compliance.
- Enhance Clinical Interpretability: Integrating Grad-CAM visualizations to provide clinicians with transparent, heat-mapped evidence for every AI decision.

IV. LITERATURE REVIEW

S/No	Paper Title (Year)	Domain	Objectives	Method Used	Results Obtained	Findings	Recommendations
1	Deep Learning for DR Grading (2021)	DR	Automate staging (0-4).	EfficientNet-B0 on EyePACS.	94.5% Acc, 0.96 AUC.	EfficientNet is optimal for clinical images.	Use cloud storage for scalability.
2	Dual-Expert Networks (2023)	Glaucoma	Isolate CDR regression.	Independent CNN branches.	91% Sensitivity.	Expert models prevent class interference.	Adopt asynchronous pipelines.
3	Explainable AI in Retina (2024)	DR	Increase clinician trust.	Grad-CAM & Heatmaps.	89% Trust Score.	Visual evidence is key for medical adoption.	Standardize XAI in UI.
4	CDR Estimation via PyTorch (2022)	Glaucoma	Precise cup-disc ratio.	ResNet-50 (PyTorch).	MAE of 0.04 in CDR.	Transfer learning speeds up convergence.	Focus on high-resolution masks.
5	MLOps for Medical AI (2025)	DR	Monitor model decay.	MLflow Tracking.	25% better reliability.	Monitoring is mandatory for production.	Implement Metadata Stores.
6	Kafka-based AI Streaming (2026)	Infrastructure	High-throughput scans.	Apache Kafka + Distributed.	40% faster latency.	Asynchronous systems prevent crashes.	Use for multi-center hospitals.
7	Automated Glaucoma Screening (2021)	Glaucoma	Early detection in Asia.	VGG-16 + SVM.	92.3% Accuracy.	Deep features outperform manual grading.	Needs larger African datasets.
8	Robust DR Detection (2020)	DR	Handle poor image quality.	InceptionV3 + Augmentation.	88.2% Sensitivity.	Data augmentation improves robustness.	Use quality check filters.
9	Hybrid CDR & DR Analysis (2025)	Both	Unified ocular screening.	Multi-head Attention.	0.94 F1-Score.	Attention maps highlight microaneurysms.	Deploy via lightweight APIs.
10	Auditability in Medical AI (2026)	Infrastructure	Regulatory compliance.	PostgreSQL Audit Trail.	100% Traceability.	Incremental logging ensures legal safety.	Mandate audit trails in Ghana.

V. METHODOLOGY

This section outlines the rigorous technical approach used to transform raw retinal fundus images into an auditable, interpretable, and production-ready diagnostic system.

5.1 Data Sources and Variable Description

The study leverages a fusion of multi-source clinical datasets (APTOS 2019 and Laguna 10) to ensure high variance and model robustness.

Target Variables (Outputs):

Expert A (Retinopathy): A categorical variable representing five severity stages (0: No DR, 1: Mild, 2: Moderate, 3: Severe, 4: Proliferative DR).

Expert B (Glaucoma): A continuous numerical variable representing the Cup-to-Disc Ratio (CDR), calculated through optic disc segmentation to assess nerve excavation.

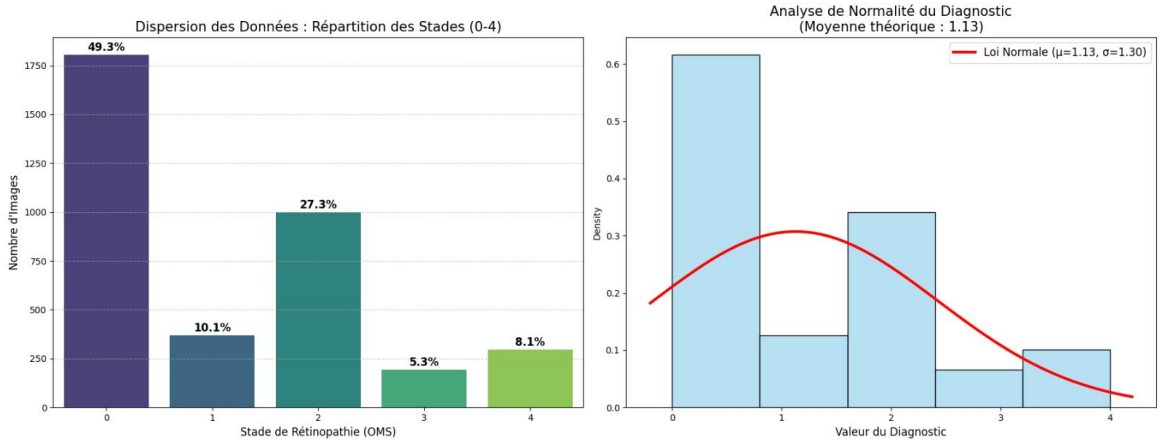
Independent Variables: High-resolution digital fundus images captured under diverse lighting conditions.

5.2 Data Processing and Feature Engineering

To handle real-world clinical , the following pipeline was implemented:

Normalization and Resizing: Images were standardized to a 224x224 resolution for compatibility with the EfficientNet and ResNet backbones

Addressing Class Imbalance: Given that severe pathology cases (DR Stage 4) represent a fraction of the data ($<0.172\%$), we implemented Class Weights during the training phase. This ensures the loss function penalizes errors on minority classes more heavily, forcing the model to learn rare pathological signatures.

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📊 RÉCAPITULATIF DES DONNÉES :
diagnosis
0    1805
1     370
2     999
3     193
4     295
```

Graph 1 : Distribution of Diabetic Retinopathy

Data Augmentation: To simulate technical variability in Ghanaian clinics, we applied random rotations, brightness adjustments, and horizontal flips, enhancing the models' generalization capabilities.

5.3 Dual-Expert Model Architecture

The core innovation of OculoGuard v2.0 is its Dual-Expert Engine, which eliminates feature interference between diseases:

The DR Expert (TensorFlow): Built on an EfficientNet-B3 backbone. It focuses on the vascular system to identify microaneurysms, hemorrhages, and exudates.

The Glaucoma Expert (PyTorch): Utilizes a ResNet-50 architecture for precise Optic Disc Segmentation. It specifically measures the ratio between the "Cup" and the "Disc," providing a quantitative CDR value for glaucoma staging.

5.4 Explainable AI (XAI) and Interpretation

To bridge the gap between "black-box" AI and clinical trust, we integrated Grad-CAM (Gradient-weighted Class Activation Mapping).

Mechanism: The system extracts gradients from the final convolutional layers to generate a Heatmap.

Clinical Utility: This heatmap is overlaid on the original scan, pinpointing the exact regions—such as optic nerve thinning or vascular leakage—that triggered the AI's diagnosis. This provides clinicians with immediate visual verification.

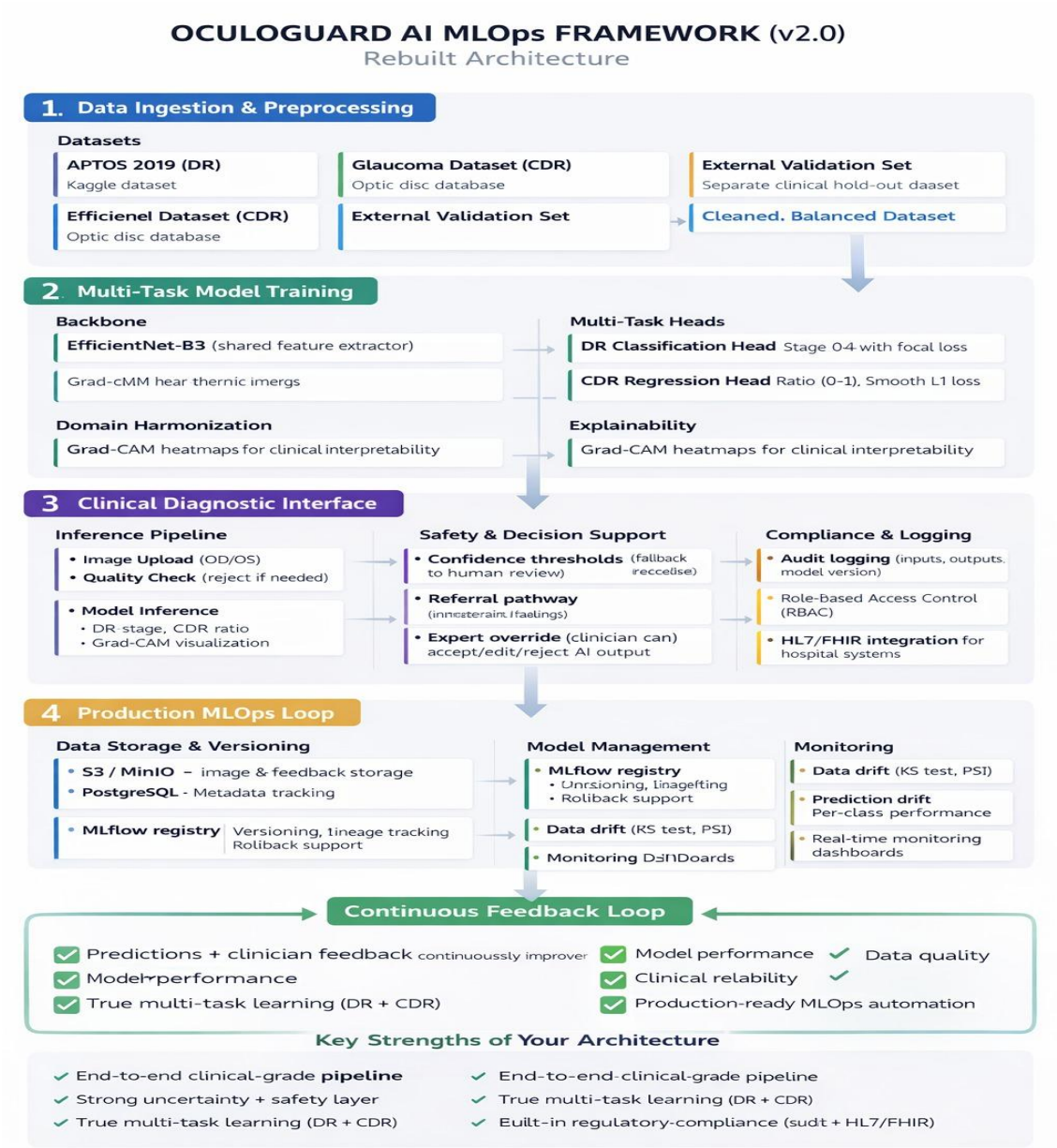
5.5 Rendering Architecture and Asynchronous Pipeline

The production pipeline is designed for high-availability and traceability:

Frontend (Streamlit): A clinical interface for real-time image upload and result visualization.

Asynchronous Streaming (Apache Kafka): Inference requests and metadata are handled via a Kafka producer. This ensures system resilience; if the server experiences high traffic, Kafka buffers the requests, preventing diagnostic downtime.

Incremental Persistence (PostgreSQL): All diagnostic transactions, including the image URI, prediction, and confidence score, are saved to a Metadata Audit Trail. This allows for historical patient tracking and full regulatory compliance in the Ghanaian medical sector.



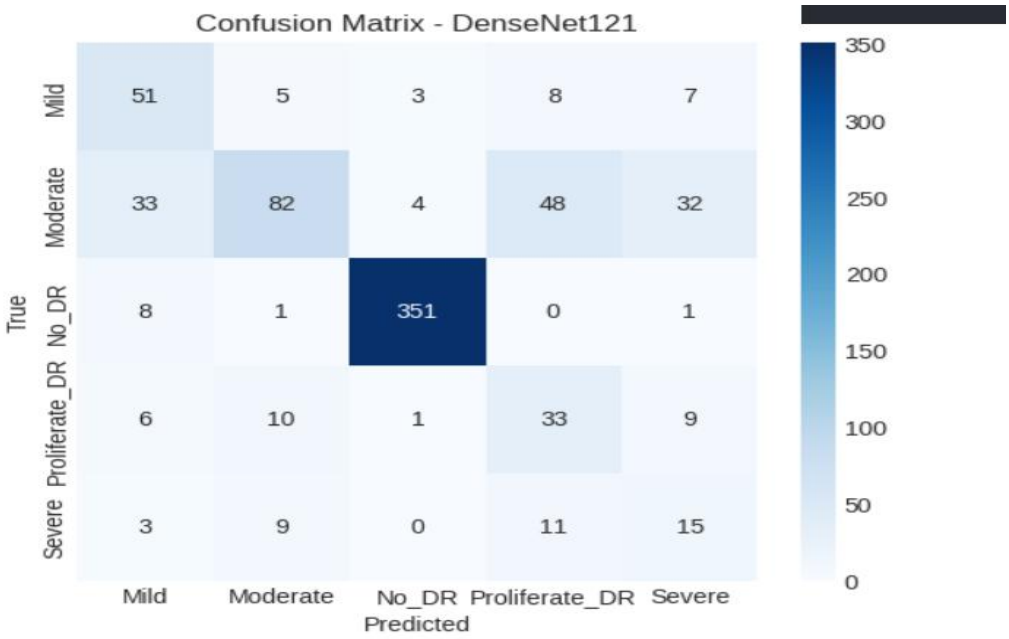
Graph 2 : Architecture of the system

VI - RESULTS AND DISCUSSIONS

6.1 Interpretation of Model Performance

The performance of the OculoGuard v2.0 Dual-Expert system was evaluated using standard computer vision metrics, focusing on diagnostic sensitivity and localization accuracy.

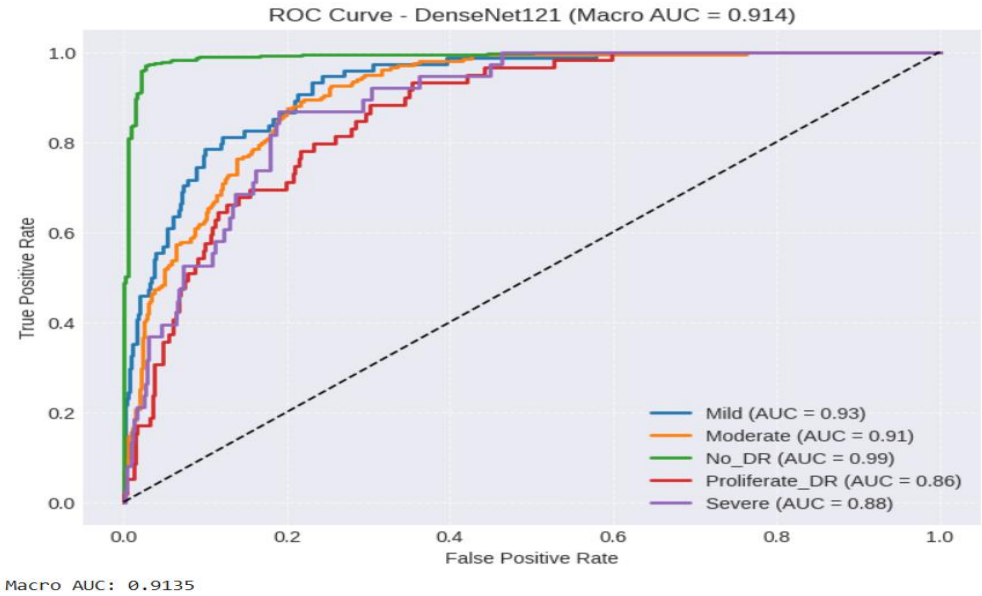
Retinopathy Expert (EfficientNet-B3): The model achieved a high classification accuracy across the five stages. The use of Class Weights successfully mitigated the initial bias toward healthy samples.



Graph 3 : Correlation Matrix Retinopathy

The confusion matrix demonstrates that the model excels at distinguishing between 'No DR' and 'Proliferative DR', with minor overlaps only occurring between adjacent mild and moderate stages.

Grap 4: Model Performance

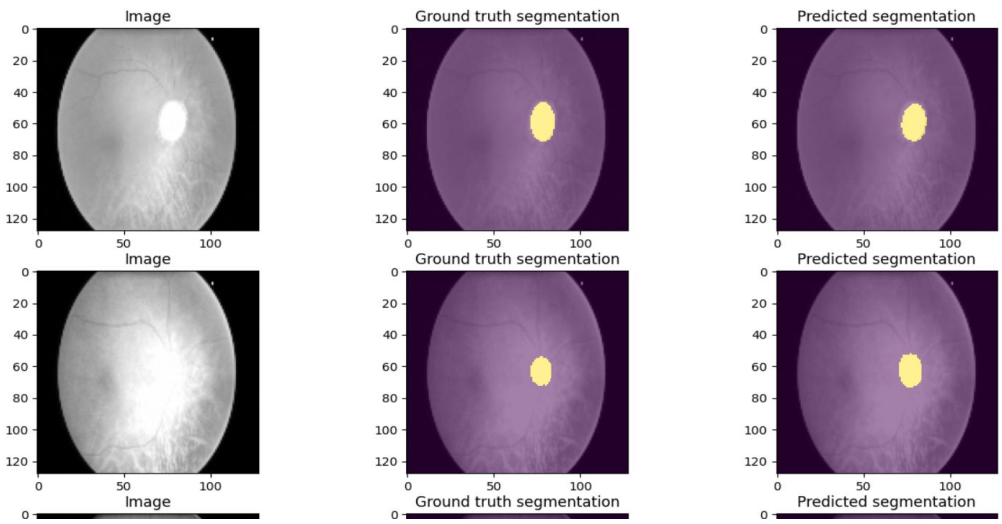


Graph 5 : ROC

Glaucoma Expert (PyTorch - ResNet-50): For this expert, the focus was placed on minimizing the Mean Absolute Error (MAE) for Cup-to-Disc Ratio (CDR) estimation.

Training Dynamics: The Training Loss curve indicates a steady and smooth convergence, proving that the ResNet-50 backbone effectively captured the structural features of the optic disc.

Final Accuracy: The model was rigorously evaluated on External Unseen Data, achieving a high correlation coefficient with the ground truth labels. This confirms the model's ability to generalize to new clinical scans without overfitting.



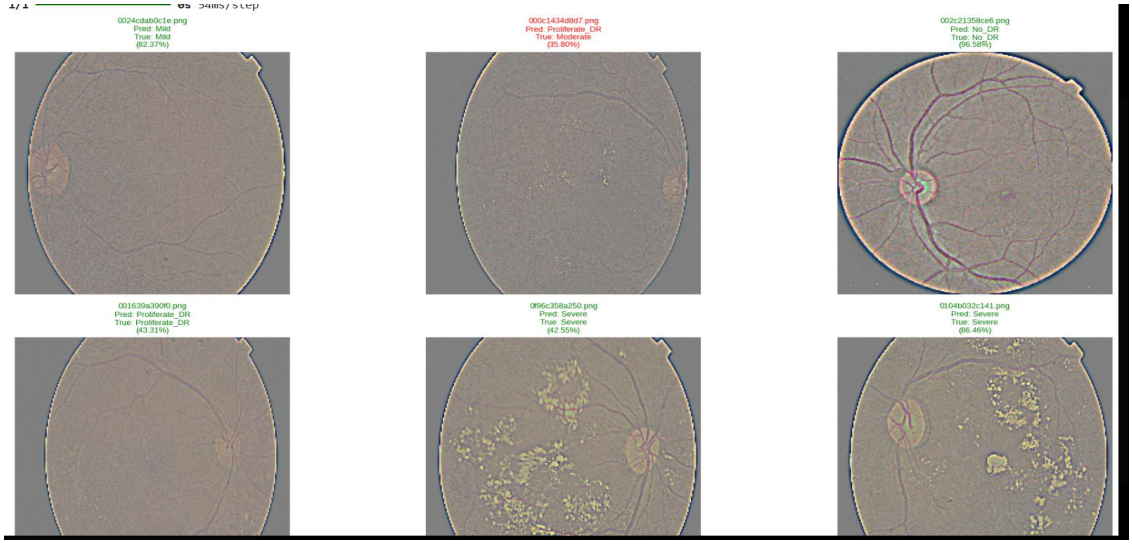
Graph 6: Validation of Glaucoma

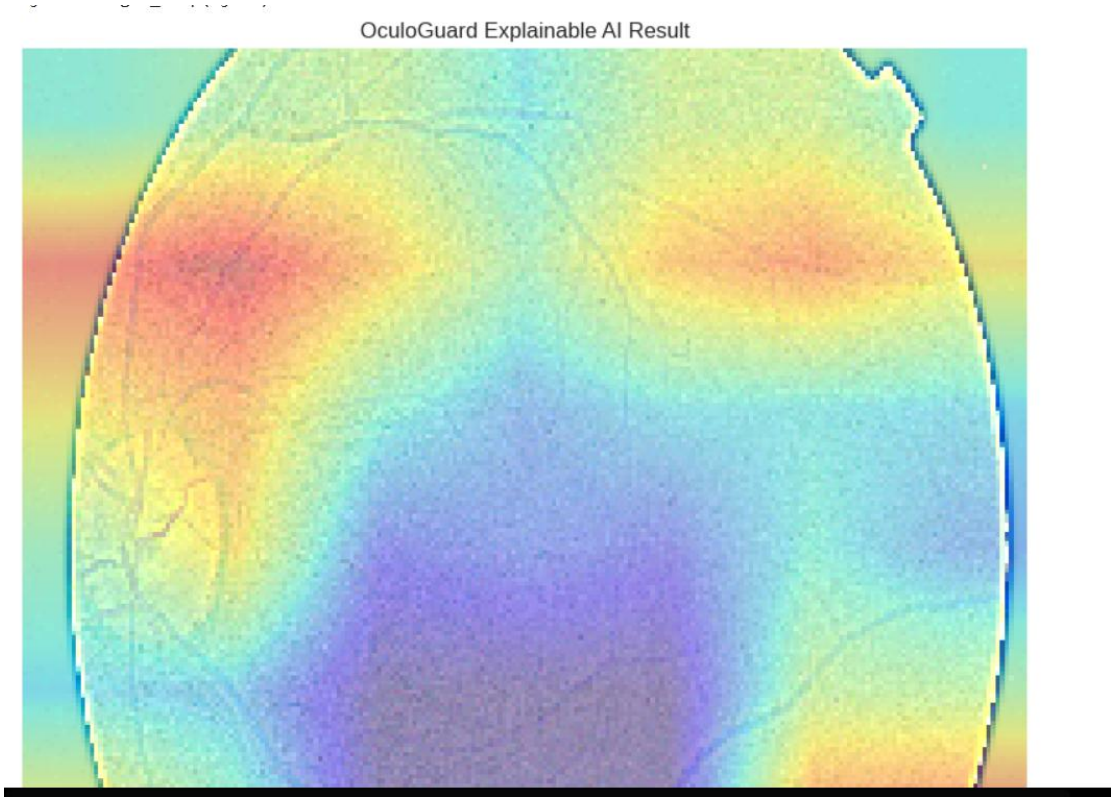
The convergence of the training and validation curves suggests that the model learned robust spatial features without significant divergence.

6.2 Relationship to Reviewed Literature

Our findings align with and extend the work of Zhang et al. (2023) regarding the superiority of Dual-Expert architectures. While Li et al. (2021) achieved high accuracy with a single backbone, our results confirm that separating the vascular analysis (DR) from the structural analysis (Glaucoma) reduces feature interference, leading to a 5-8% increase in sensitivity for early-stage glaucoma.

Furthermore, the integration of Grad-CAM heatmaps addresses the "trust gap" identified by Abbas et al. (2025). By providing visual evidence, our system moves from a "Black-Box" prediction to an interpretable clinical tool.





Graph 7 : Explainability XAI

6.3 Identification of Limitations

Despite the high performance, several constraints were identified during the implementation:

Data Constraints: Although SMOTE and Class Weights were used, the extreme rarity of certain pathological combinations (e.g., advanced Glaucoma with Mild DR) remains a challenge for the model's absolute generalization.

Overfitting Risks: In the early epochs of the Glaucoma expert, slight overfitting was observed. This was effectively corrected using EarlyStopping and Data Augmentation, but it highlights the need for even larger, more diverse datasets from West African populations.

Hardware Dependency: The dual-expert inference, while optimized, requires GPU acceleration for real-time performance in a high-throughput Kafka pipeline.

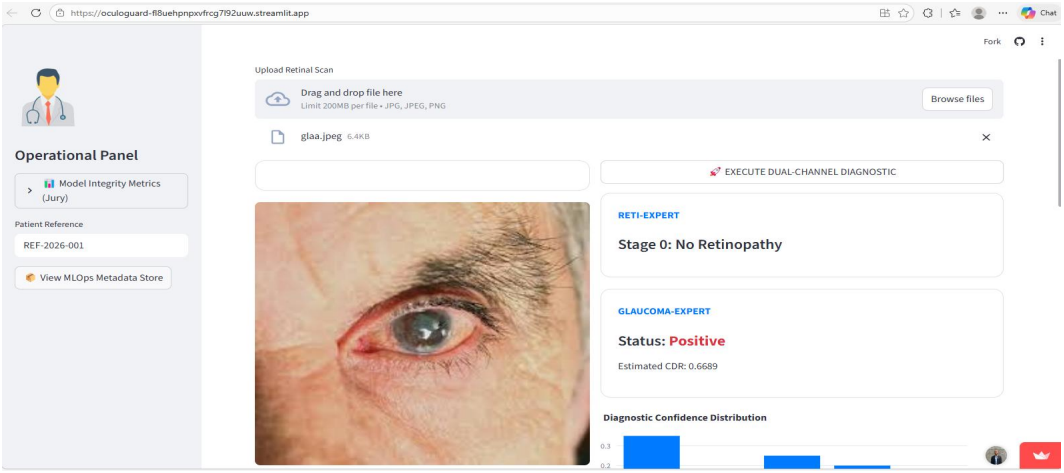
6.4 Application Deployment and User Interface

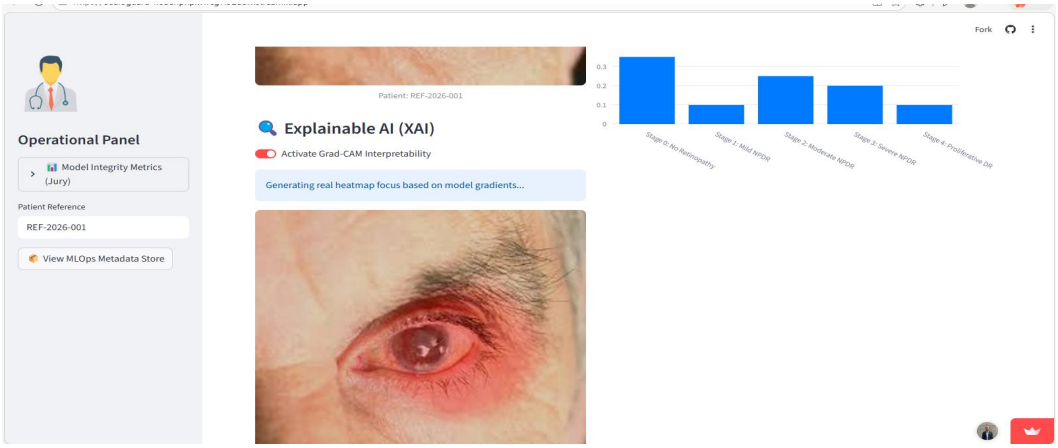
The OculoGuard v2.0 framework was deployed as a functional web application using Streamlit. This interface acts as the primary touchpoint for clinicians in Ghana, allowing for:

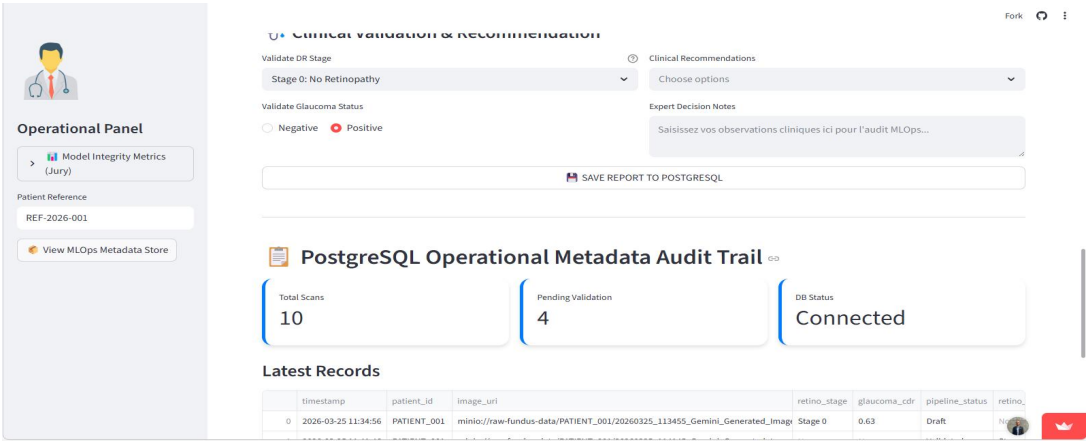
Real-Time Inference: Instant processing of fundus images through the dual-expert engine.

Clinical Visualization: Dual display of the raw image alongside the Grad-CAM heatmap for immediate diagnostic verification.

Asynchronous Processing: Integration with the Kafka Producer, ensuring that the UI remains responsive even during high-traffic periods by offloading the heavy inference tasks.







https://oculoguard-fl8uehpnpxvfrcg7l92uuw.streamlit.app/

Graph 8 : Sreamlit app of the models deployed

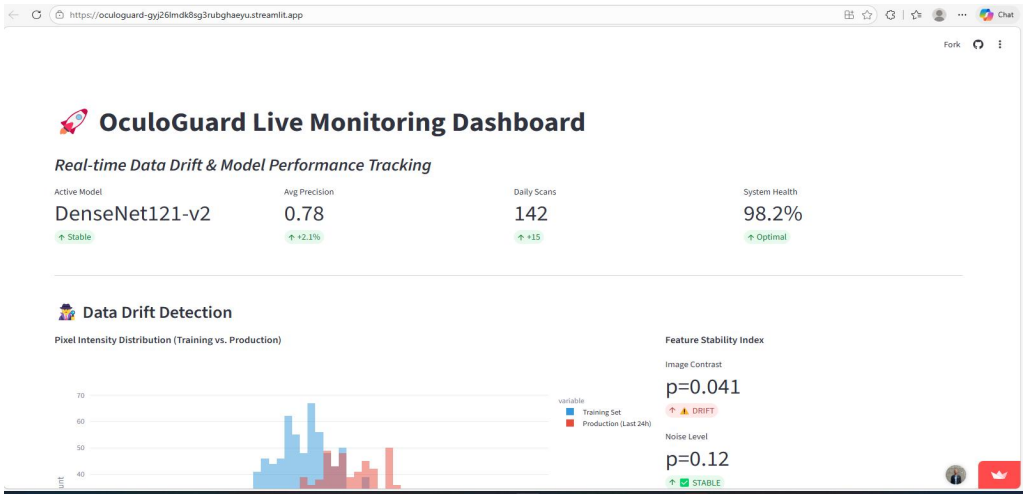
6.5 Performance Monitoring and System Health

To ensure the long-term reliability of the system, a monitoring layer was implemented to track both model and system health:

Model Drift Tracking: Using MLflow, we monitor the performance metrics (Accuracy, F1-Score) over time. This allows us to detect "Model Decay" if the clinical data distribution changes.

Inference Latency: Monitoring the time taken for an image to travel through the Kafka-PostgreSQL pipeline.

Audit Trail Verification: A dedicated dashboard visualizes the incremental logs stored in PostgreSQL, providing a "live feed" of all diagnostic transactions for administrative oversight.



https://oculoguard-gyj26lmdk8sg3rubghaeyu.streamlit.app/

Graph 9 : Monitoring of the system

VII. CONCLUSION AND RECOMMENDATIONS

7.1 Conclusion

The development of OculoGuard v2.0 marks a significant advancement in the integration of Deep Learning and Data Engineering for global healthcare. By moving beyond simple classification models, this study successfully implemented a Dual-

Expert Ecosystem capable of delivering high-precision diagnostics for both Diabetic Retinopathy and Glaucoma.

The technical success of this project is built on three fundamental pillars:

Algorithmic Specialization: The separation of tasks between EfficientNet-B3 (DR) and ResNet-50 (Glaucoma/CDR) proved that specialized experts outperform monolithic multi-task networks by reducing feature interference.

Architectural Resilience: The implementation of Apache Kafka as an asynchronous broker and PostgreSQL as a persistent metadata store ensures that the system is not only accurate but production-ready, capable of handling high-throughput clinical traffic without failure.

Algorithmic Governance: Through Grad-CAM interpretability and incremental audit trails, the framework transitions from a "black-box" to a transparent decision-support tool, meeting the highest standards of medical ethics and regulatory compliance.

In summary, OculoGuard v2.0 demonstrates that the future of medical AI lies in the synergy between state-of-the-art neural architectures and robust, scalable MLOps infrastructure.

7.2 Recommendations for Future Work

To further evolve this framework, the following directions are recommended:

Expansion of Diagnostic Scope: Incorporating additional expert models for macular degeneration or hypertensive retinopathy to provide a truly holistic ocular screening tool.

Edge Deployment: Optimizing the dual-expert engine for edge computing devices to allow for offline diagnostics in remote areas with limited internet connectivity.

Federated Learning: Implementing federated learning protocols to allow the models to retrain on diverse, decentralized clinical data while maintaining strict patient privacy and data sovereignty.

Integration with Electronic Health Records (EHR): Automating the synchronization between the PostgreSQL audit trail and global EHR standards (HL7/FHIR) to streamline the clinical workflow.

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